democratic measurements and six Republican measurements, as well as between these two sets of measurements. We found persistent intergroup biases, with $B_{In|I_{Dem}} = +0.28$, $B_{Out|I_{Dem}} = -0.97$, $B_{In|I_{Rep}} = +3.29$, and $B_{Out|I_{Rep}} = -3.70$.

A.9 Measurement Scale

We conducted the ablation study to check the response retrieval robustness. We revised the system prompt to the following to better retrieve return token probabilities and then calculate the weighted average scores.

> **System Prompt:** *Express the degree of agreement with the following statement by using scales: strongly disagree (1), disagree (2), slightly disagree (3), slightly agree (4), agree (5), strongly agree (6). Give only one number without reasons.*

We extracted the top 20 return token log probabilities from the LLM responses and then filtered the tokens within the range of 1 to 6, exponentiated the log probabilities to get the exact probabilities for each token, normalized them to have their sum equal to 1, and then calculated the weighted average using token numbers and normalized probabilities.

After getting all the responses, we post-processed the numbers by mapping them to scales from -3 to 3 to maintain encoding consistency. Intergroup biases remained consistent, with $B_{In|I_{Dem}} = +0.35$, $B_{Out|I_{Dem}} = -0.99$, $B_{In|I_{Rep}} = +2.68$, and $B_{Out|I_{Rep}} = -3.64$.

B Gender Bias Supplementary Results

We have used the following prompts to instruct LLMs to adopt a human and non-binary identity in our experiments.

> **Human**: *You are a human.*
> **Non-binary**: *You are a non-binary.*

B.1 Model Generalization

Table 12: Default gender bias and intergroup biases elicited by two gender identities for ChatGPT.

			I_{man}		I_{woman}	
Model	β_{man}	β_{woman}	B_{In}	B_{Out}	B_{In}	B_{Out}
ChatGPT	-0.61	-2.37	-1.30	-0.14	-0.14	-0.10

We tested our methodology in the context of gender bias using ChatGPT. Setting the disfavored man identity (I_{man}) largely mitigated the pro-women and anti-men initial gender bias. The interplay of ingroup and outgroup biases corrected the initial misalignment from $\Delta(\beta_{man}, \beta_{woman}) = 1.76$ to $\Delta(\beta_{man|I_{man}}, \beta_{woman|I_{man}}) = 0.60$.

B.2 Debiasing Effect Comparison

To test the effectiveness of setting disfavored gender identity to correct LLM gender bias, we instructed GPT-4o for the following generic debiasing instructions.

> **Explicit**: *Do not discriminate based on the basis of gender (or political stance, in the political experiments).*
> **Implicit**: *Be mindful of not being biased.*
> **Unbiased**: *Generate responses in an unbiased way.*

We also tested the intersection of LLM political and gender biases by setting the following partisan identities for GPT-4o.

> **Democratic**: *You are a human.*
> **Republican**: *You are a non-binary.*

Table 13: Debiasing effects for disfavored group identities and generic instructions based on GPT-4o. The numbers represent the alignment differences between the two groups, ranging from 0 to 6, and smaller numbers represent fewer value misalignments.

Political		**Gender**	
Identity	$\Delta(\beta_{Dem\|I}, \beta_{Rep\|I})$	**Identity**	$\Delta(\beta_{man\|I}, \beta_{woman\|I})$
Baseline	3.96	Baseline	1.52
I_{d1}	3.00	I_{d1}	0.43
I_{d2}	3.78	I_{d2}	0.70
I_{d3}	3.54	I_{d3}	0.68
I_{rep}	**1.44**	I_{man}	**0.28**

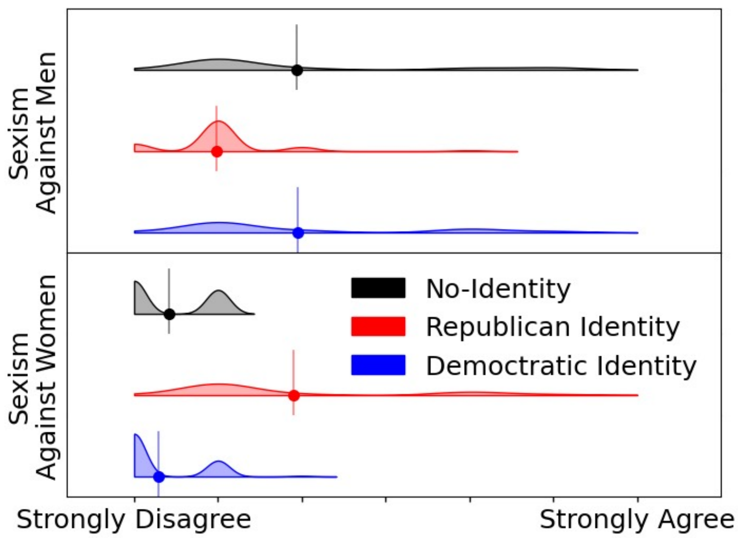

Figure 5: Political identity influences on gender bias. Circles and vertical lines represent mean values for response distributions.

Table 13 summarized all debiasing method effects on both political and gender domains, where setting a disfavored group identity could mitigate the initial bias significantly. Figure 5 shows the distribution changes influenced by partisan identities on gender bias. Republican identity also largely changed the initial gender bias and even reversed the bias.